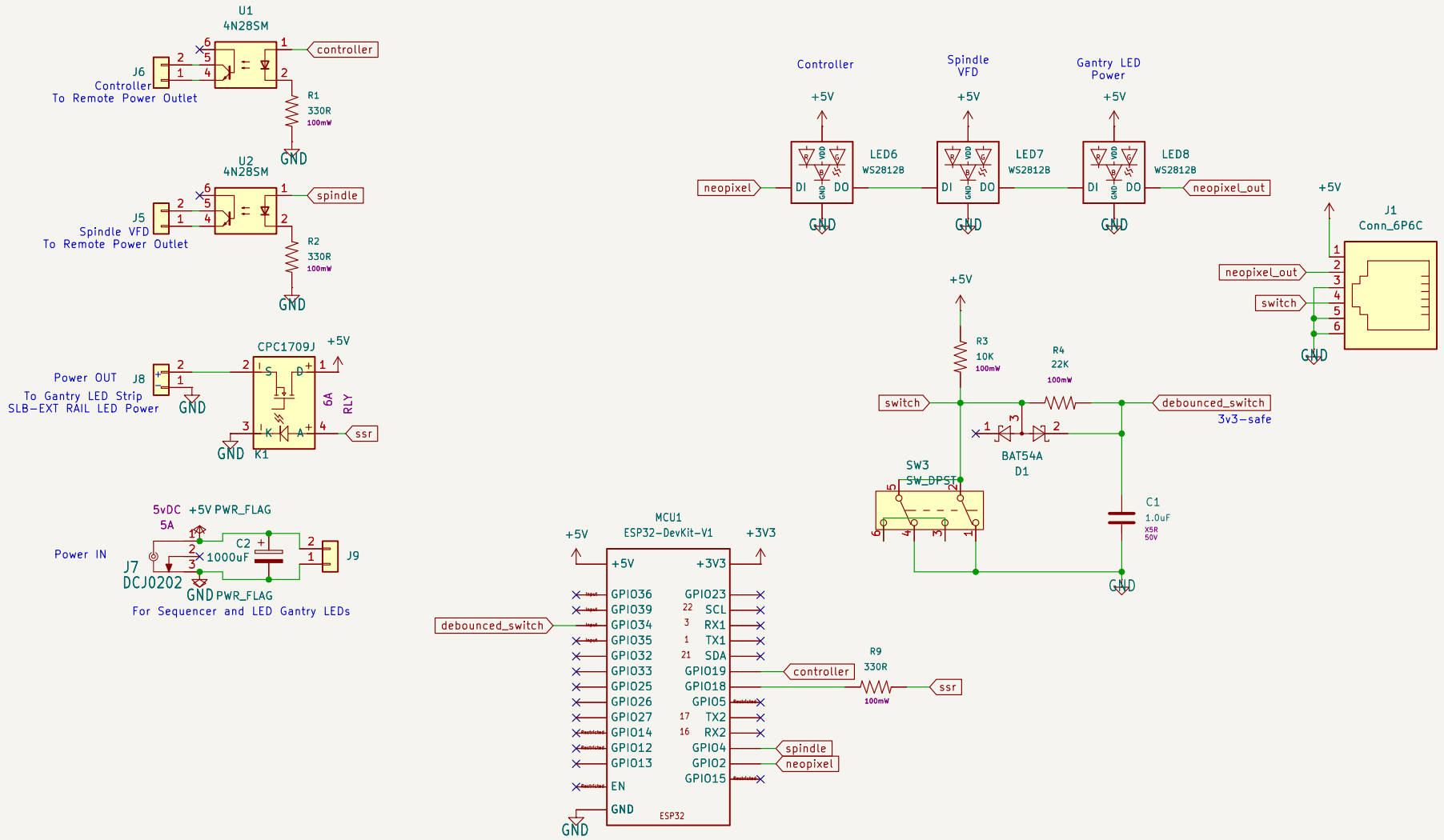
U1
4N28SM
J6
Controller
To Remote Power Outlet
controller
R1
330R
100mW
GND
U2
4N28SM
J5
Spindle VFD
To Remote Power Outlet
spindle
R2
330R
100mW
GND
CPC1709J
+5V
Power OUT
J8
To Gantry LED Strip
SLB-EXT RAIL LED Power
GND
6A
RLY
ssr
GND
K1
5vDC
5A
+5V PWR_FLAG
Power IN
J7
DCJ0202
C2
1000uF
J9
GND PWR_FLAG
For Sequencer and LED Gantry LEDs
Controller
Spindle VFD
Gantry LED Power
+5V
LED6
LED7
LED8
WS2812B
neopixel
neopixel_out
DI
DO
VDD
GND
J1
Conn_6P6C
switch
R3
10K
100mW
R4
22K
100mW
debounced_switch
3v3-safe
BAT54A
D1
SW3
SW_DPST
C1
1.0uF
X5R
50V
MCU1
ESP32-DevKit-V1
+3V3
debounced_switch
GPIO36
GPIO39
GPIO34
GPIO35
GPIO32
GPIO33
GPIO25
GPIO26
GPIO27
GPIO14
GPIO12
GPIO13
EN
GPIO23
SCL
RX1
TX1
SDA
GPIO19
GPIO18
GPIO5
TX2
RX2
GPIO4
GPIO2
GPIO15
R9
330R
100mW
spindle
neopixel
ESP32
SPCoast
Board Boilerplate
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Title: AltmillSwitchController
John Plocher
Size: USLetter
Date: 2026.01
Rev: 1.0
KiCad E.D.A. 10.0.1
Id: 1/1
active